

Technical Report

Urban Heat Island (UHI) Spatio-Temporal Analysis and Deep Learning LULC Classification (Bokaro: 2015-2024)

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Abstract

This study presents a multi-model framework for decadal Land Use/Land Cover (LULC) change detection and Urban Heat Island (UHI) quantification in Bokaro district of Jharkhand (2015–2024). Utilizing Landsat 8 surface reflectance composites, a feature-decoupling strategy was employed to mitigate temporal domain shift. A nine-feature spectral stack comprising six optical bands and three derived indices (*NDVI*, *NDBI*, *MNDWI*) was used to train epoch-specific Random Forest classifiers, deliberately excluding Land Surface Temperature (LST) parameters.

Spatially independent validation yielded an overall accuracy of 90.1% for 2015 and 98.5% for 2024. Spatial analyses indicate a 7.9% expansion in built-up area and a de-trended LULC-attributable surface warming of 5.26°C over the decade. Post-hoc mechanistic interpretability via SHAP attribution confirms that deep learning architectures natively exploit shortwave infrared (SWIR) reflectance contrasts, effectively validating the physical grounding of the classification logic. Local multi-model benchmarking confirms that classification performance remains robust and feature-driven across varying algorithmic architectures.

Introduction

Rapid urbanisation in developing economies induces systematic modifications of the surface energy balance. The progressive replacement of vegetated, pervious land with impervious built surfaces—concrete, asphalt, industrial rooftops—reduces evaporative cooling, increases sensible heat storage, and diminishes longwave radiative loss at night. These combined effects produce the Urban Heat Island (UHI) phenomenon: the measurable thermal excess of urban cores relative to their rural surrounds. Over multi-decadal timescales, UHI intensification poses compounding risks to public health, energy demand, and ecosystem function.

India's industrial cities present a particularly instructive setting for studying these dynamics. Bokaro Steel City, established in the 1960s as one of India's flagship public-sector steel towns, combines a dense industrial footprint with ongoing residential and infrastructural expansion. The surrounding Damodar River basin and dry deciduous forest tracts provide thermal contrast against which urban warming can be measured. Despite its industrial significance and documented post-2000 growth, Bokaro remains understudied in the peer-reviewed remote sensing literature on UHI.

Satellite-based Land Surface Temperature (LST) retrieval, combined with multi-temporal Land Use / Land Cover (LULC) classification, provides a scalable, cost-effective means of monitoring urban thermal evolution. Landsat's thirty-year multispectral archive at 30 m spatial resolution makes it the primary data source for decadal urban studies. However, multi-epoch analysis introduces a non-trivial methodological complication: temporal domain shift. Inter-annual variation in atmospheric composition, soil moisture, phenological state, and sensor calibration

causes systematic spectral drift between epochs. A classifier trained on one epoch's absolute values can fail when applied to another, producing physically implausible class assignments.

This study addresses these challenges through an end-to-end, reproducible framework implemented across two compute environments: Google Earth Engine for cloud-based preprocessing and epoch-specific classification, and a local Python environment for multi-model benchmarking, scaled spatial inference, and post-hoc interpretability analysis. The central methodological contribution is a feature-decoupling strategy that eliminates absolute thermal signals from the classification feature space, forcing models to rely on structurally stable spectral ratios and thereby suppressing cross-epoch phantom-class artifacts.

Objectives

The study pursues three specific aims:

- **Cloud-Based Temporal LULC Mapping:** Construct cloud-free, pre-monsoon Landsat 8 composites for 2015 and 2024 within GEE and produce epoch-specific LULC classifications using spectrally decoupled Random Forest models.
- **Multi-Model Benchmarking:** Validate GEE-derived features against three locally trained classifiers (RF, XGBoost, PyTorch MLP) to assess whether classification performance is architecture-dependent or feature-dependent.
- **Mechanistic Interpretability:** Apply SHAP attribution to determine whether models align with physically grounded spectral reasoning rather than statistical memorization of training artifacts.

Study Area

The study domain is the Bokaro administrative district in Jharkhand, eastern India (approximately 23°29'–23°54'N, 85°42'–86°26'E). The district encompasses roughly 2,861 km² of topographically varied terrain, including the Chota Nagpur Plateau upland, the Damodar River valley, and the built-up core of Bokaro Steel City.

Bokaro Steel City was established in 1964 as a planned industrial township anchored by the Bokaro Steel Plant (BSP), one of the largest integrated steel facilities in Asia. The urban built environment is structurally characterised by expansive industrial structures, slag dump terraces, wide-grid road networks, and high-density residential sectors originally designed for steel plant employees.

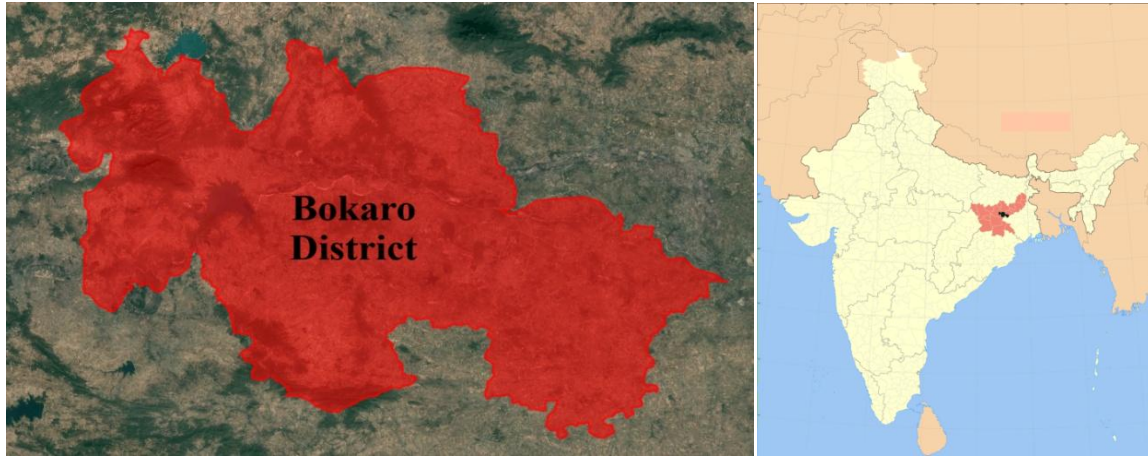


Figure 1a (left): *Bokaro district satellite view from Google Earth; Figure 1b (right):* *Bokaro, Jharkhand in map of India*

Surrounding the core are semi-urban residential zones, agricultural tracts, open-cast coal and mineral extraction sites, and fragments of tropical dry deciduous forest—producing sharp land cover contrasts that make the region well-suited for spectral LULC discrimination.

The district falls within a humid subtropical climate zone transitioning towards tropical wet-and-dry (Köppen Aw). Pre-monsoon months (March–May) are characterised by extreme heat and low humidity, with surface temperatures frequently exceeding 40°C. This thermal extreme makes the pre-monsoon window the optimal acquisition period for UHI analysis: vegetation stress reduces canopy cooling, soil moisture is at annual minima, and the thermal contrast between impervious and natural surfaces is at its widest. These properties maximise the signal-to-noise ratio for surface temperature retrieval and built-up area discrimination.

Data Preprocessing

Satellite Data

The primary data source is Landsat 8 Operational Land Imager / Thermal Infrared Sensor (OLI/TIRS) Collection 2, Tier 1, Level-2 Surface Reflectance (LANDSAT/LC08/C02/T1_L2), accessed via the GEE public catalogue. Collection 2 Level-2 products provide atmospherically corrected, scaled surface reflectance values with associated quality assessment (QA_PIXEL) bands, making them suitable for multi-temporal analysis without additional atmospheric correction.

Two seasonal composites were constructed: a baseline epoch spanning March 1–May 31, 2015, and a target epoch spanning March 1–May 31, 2024. Within each window, all available Landsat 8 scenes intersecting the Bokaro FAO-GAUL Level-2 boundary were collected. Compositing used pixel-wise median aggregation, which suppresses residual cloud and shadow contamination while preserving the central tendency of clear-sky observations.

Cloud and Quality Masking

Scene-level pre-filtering was applied prior to pixel-level masking: only scenes with a reported CLOUD_COVER metadata value below 15% were admitted to each collection using `ee.Filter.lt('CLOUD_COVER', 15)`. This upstream filter substantially reduces the number of scenes entering the median composite and guards against persistent cloud contamination that pixel-level CFMask flags may miss in individual acquisitions. Following scene selection, residual cloud and cloud-shadow were removed per-pixel using QA_PIXEL bit flags at positions 3 (cloud) and 4 (cloud shadow). Masked images were clipped to the district boundary before median reduction.

A remaining limitation is that CFMask can misclassify bright industrial surfaces—slag deposits and high-albedo concrete rooftops—as cloud in individual scenes. The scene-level 15% threshold reduces but does not eliminate this risk, as it is based on the fraction of cloudy pixels reported in metadata, which is itself derived from the same algorithm.

Spectral Band Calibration

Optical bands (SR_B2 through SR_B7) were rescaled to surface reflectance using the Collection 2 scale and offset coefficients: $\rho = \text{DN} \times 0.0000275 - 0.2$. The thermal band (ST_B10) was converted to brightness temperature in Kelvin using the corresponding scale factor (0.00341802) and offset (149.0 K).

Feature Engineering

Spectral Indices

Three spectral indices were derived from the calibrated reflectance bands to mathematically isolate the primary LULC classes of interest.

The *Normalised Difference Vegetation Index (NDVI)* exploits the spectral contrast between near-infrared (NIR) reflectance, where healthy chlorophyll-bearing canopies are strongly reflective, and red reflectance, where they are absorptive:

$$NDVI = (SR_B5 - SR_B4) / (SR_B5 + SR_B4)$$

The *Normalised Difference Built-Up Index (NDBI)* targets the reflectance contrast between shortwave infrared (SWIR1) and NIR, where impervious materials exhibit elevated SWIR reflectance and suppressed NIR relative to vegetation:

$$NDBI = (SR_B6 - SR_B5) / (SR_B6 + SR_B5)$$

The *Modified Normalized Difference Water Index (MNDWI)*, following Xu (2006), substitutes SWIR1 for the NIR band used in the original NDWI formulation. This substitution suppresses built-up reflectance contributions, which can contaminate NIR-based water indices in urban settings:

$$MNDWI = (SR_B3 - SR_B6) / (SR_B3 + SR_B6)$$

The final nine-feature input vector fed to all classifiers comprised: SR_B2, SR_B3, SR_B4, SR_B5, SR_B6, SR_B7, NDVI, NDBI, and MNDWI. Land Surface Temperature was deliberately excluded from this feature set for reasons detailed in the following sections.

Land Surface Temperature Derivation

LST was derived separately for thermal mapping and change analysis, but not used as a classification input. Fractional Vegetation Cover (FVC) was estimated from NDVI using empirically constrained minimum and maximum thresholds (NDVImin = 0.1, NDVImax = 0.6):

$$FVC = [(NDVI - NDVImin) / (NDVImax - NDVImin)]^2$$

Land Surface Emissivity (LSE) was computed by a simplified two-endmember model:

$$LSE = 0.004 \times FVC + 0.986$$

This formulation represents a linearised approximation of the more general expression $LSE = \epsilon_v \times FVC + \epsilon_s \times (1 - FVC) + C$, where C accounts for surface roughness contributions. In the GEE implementation, the constants were folded into a single linear expression (slope 0.004, intercept 0.986), which implicitly sets $\epsilon_s \approx 0.986$ and $\epsilon_v - \epsilon_s \approx 0.004$. This is a simplification relative to the literature-standard values ($\epsilon_s \approx 0.97$, $\epsilon_v \approx 0.99$, $C \approx 0.005$), and constitutes a recognised limitation of the LST retrieval.

Absolute LST in degrees Celsius was derived from brightness temperature (BT in Kelvin) by correcting for emissivity:

$$LST = BT / [1 + (0.00115 \times BT / 1.4388) \times \ln(LSE)] - 273.15$$

This expression follows the Planck function inversion approach described in Sobrino et al. (2004). The constant 0.00115 approximates the effective wavelength of TIRS Band 10 ($\lambda \approx 10.89 \mu m$) and $1.4388 \times 10^{-2} m \cdot K$ is the second radiation constant (hc/k). The GEE implementation applied this formula using the ST_B10 scaled product (in Kelvin) directly as BT without first converting to apparent radiance—a procedural detail that should be verified against the Collection 2 TIRS product specifications for absolute accuracy claims.

Temporal Domain Shift and Mitigation

The ‘Flooding’ Artifact

In the initial configuration of this study, a single Random Forest classifier was trained on the 2024 composite and applied retrospectively to the 2015 composite. This produced a systematic misclassification that is termed "flooding" artifact: highly reflective industrial surfaces—specifically, smooth concrete rooftops and slag deposits exhibiting low SWIR reflectance and moderate MNDWI values—were assigned to the water body class.

This error has a straightforward physical explanation. The 2024 training epoch was characterized by higher mean LST and subtly different atmospheric loading relative to 2015. When LST was included as a classification feature, the model learned decision boundaries anchored partly to absolute thermal values. When those boundaries were applied to the 2015 composite—where equivalent industrial surfaces exhibited lower absolute temperatures due to inter-annual meteorological variation—the model encountered feature vectors outside its training distribution. The spectral overlap between cool industrial concrete and open water in the visible-NIR-SWIR domain, combined with the thermal out-of-distribution condition, caused the classifier to default toward the nearest in-distribution class, which was water.

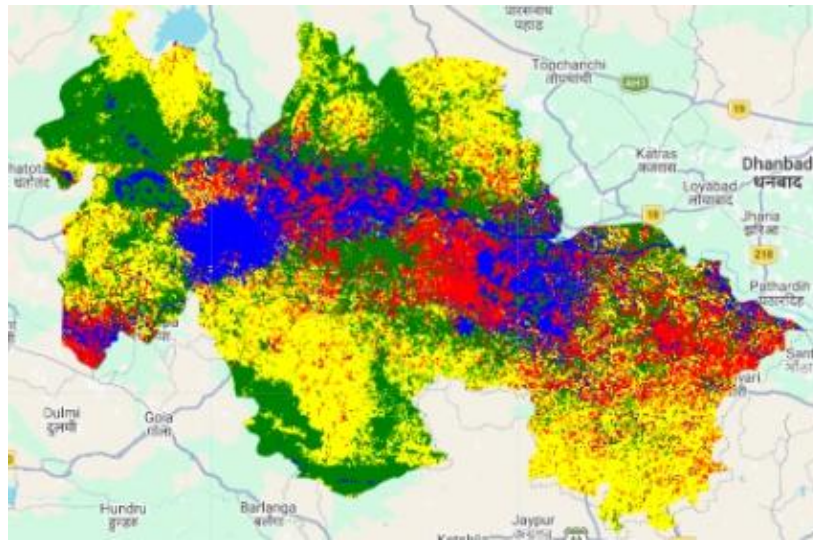


Figure 2: *A significantly large portion of the district surface gets classified as ‘water’. Here, Built-up is red, Vegetation is green, Bare Earth is yellow and water is blue.*

The domain shift was eliminated through a combination of three coordinated interventions:

- **LST Exclusion:** Absolute LST was removed from the classification feature vector entirely. All three spectral indices (NDVI, NDBI, MNDWI) are normalised ratios and therefore largely insensitive to inter-annual absolute magnitude drift, making the feature space more temporally stable.
- **MNDWI:** By retaining MNDWI as a classification input, the model gained access to a physically grounded, dimensionless discriminator between water (strongly positive MNDWI) and built-up surfaces (strongly negative MNDWI). This spectral separation is stable across years because it exploits the fundamental optical properties of water—high visible absorption in red-NIR, high SWIR absorption—rather than transient thermal signatures.
- **Epoch-Specific Classifiers:** Rather than applying a single trained classifier across both epochs, independent Random Forest models were trained on epoch-specific training polygons. Each classifier was thus calibrated to the spectral baseline of its own epoch, eliminating the need for cross-epoch feature normalization.

This three-tier strategy proved effective. Classified outputs for both 2015 and 2024 were visually coherent, with water bodies correctly identified and industrial surfaces distinguished from open water throughout the district domain.

Classification Methodology

Training Data

Training polygons were manually delineated using the GEE polygon drawing tool by myself. Being a resident of Bokaro and having the personal experience of witnessing its landscape change over the years proved to be very helpful. For the 2024 epoch, 180 polygons were drawn: 73 built-up, 37 green cover, 37 water, and 33 bare earth. For the 2015 epoch, 81 polygons were drawn: 22 built-up, 24 water, 23 bare earth, and 12 green cover. The asymmetry in sample count between epochs reflects the greater spatial complexity of the 2024 landscape and the availability of higher-resolution reference imagery for recent years. Training polygons were stored as GEE FeatureCollection assets. Each feature encodes a discrete landcover class label on a four-class schema: 0 = Built-up, 1 = Vegetation, 2 = Bare Soil / Open Land, 3 = Water.

Pixel samples were extracted from the composite image stacks at 30 m resolution using the GEE *sampleRegions()* function with *tileScale = 16* to prevent memory overflow during large polygon extraction.

Spatially Independent Train-Test Split

The train-test split was implemented at the polygon level, not the pixel level. A uniform random value was assigned to each polygon using *randomColumn()* with seed 42. Polygons with random value < 0.8 were assigned to the training set; the remainder formed the test set. Pixel samples were then extracted separately from training polygons and test polygons. This ensures that all pixels within a given polygon appear exclusively in either train or test — eliminating the spatial autocorrelation bias that arises when nearby pixels from the same polygon are split randomly across partitions. This is the methodologically correct approach for polygon-based training data, where intra-polygon pixel correlation would otherwise inflate accuracy estimates.

Separate Random Forest classifiers were then trained for 2015 and 2024, each with 50 estimators and seed = 42. The final production classifiers (used to generate the LULC maps) were trained on the full 80% polygon partition.

In-Cloud Accuracy Assessment

Accuracy assessment was performed within GEE using the withheld 20% polygon partition. The trained classifier was applied to test pixels and the resulting predictions were compared against reference labels using GEE's *errorMatrix()* function. Results for both epochs are reported below.

2015 Classifier (Overall Accuracy: 90.1% | Kappa Coefficient: 0.756)

The 2015 kappa of 0.756 indicates substantial agreement but falls notably short of the 2024 result. The primary source of error is Bare Soil, which achieved only 29% recall — of 127 bare soil test pixels, 59 were misclassified as Vegetation and 31 as Water. This reflects the limited number of bare earth training polygons in the 2015 asset (23 polygons) and the spectral similarity between dry open soil and sparse, stressed vegetation in pre-monsoon conditions. Vegetation precision is also low at 60%, meaning 255 non-vegetation pixels (mainly built-up and bare soil) were incorrectly assigned to the vegetation class.

Class	True Positive	Recall	Precision	F1
Built-up	664	82.9%	76.4%	79.5%
Vegetation	384	91.6%	60.1%	72.6%
Bare Soil	37	29.1%	26.2%	27.6%
Water	4418	92.9%	99.1%	95.9%

Table 1

2024 Classifier (Overall Accuracy: 98.5% | Kappa Coefficient: 0.934)

The 2024 classifier performs substantially better overall, driven by the much larger and better-balanced training dataset. Vegetation is near-perfectly classified (F1 = 0.994). Bare Soil remains the weakest class (F1 = 0.553): 47 of 212 test pixels were assigned to Built-up and 21 to Vegetation. The Water class shows lower recall (70.5%) than in 2015 despite the larger dataset — the 2024 water training set has only 37 polygons, fewer than the 2015 set, and water bodies at the edges of the district boundary may present spectrally ambiguous signatures under extreme pre-monsoon conditions.

Class	True Positive	Recall	Precision	F1
Built-up	4386	92.7%	97.3%	95.0%
Vegetation	33118	99.7%	99.3%	99.4%
Bare Soil	122	57.5%	53.3%	55.3%
Water	122	70.5%	68.9%	69.7%

Table 2

Meteorological Detrending

Rural Control and UHI Intensity

The canonical definition of Urban Heat Island Intensity (UHII) is the difference between urban surface temperature and a contemporaneous rural reference: $UHII = LST_Urban - LST_Rural$. Comparing only urban LST between two years conflates LULC-driven warming with background regional climate trends. To isolate the urban-specific signal, a stable rural control was constructed by masking to pixels classified as Vegetation (Class 1) in both the 2015 and 2024 LULC maps. These pixels, which remained vegetated across the full decade, represent a land-cover-neutral thermal baseline. Mean LST was extracted over this mask for both epochs.

Results: Rural control LST rose from 41.97°C (2015) to 48.28°C (2024), a change of 6.31°C. Urban (district-wide) LST rose from 42.25°C to 48.70°C, a change of 6.45°C. UHII in 2015 was 0.28°C; in 2024 it was 0.42°C — an intensification of 0.14°C over the decade.

The small absolute UHII values ($< 0.5^\circ\text{C}$ in both years) warrant honest interpretation. Bokaro's industrial landscape contains large areas of bare concrete, slag deposits, and metal rooftops that heat comparably to or hotter than the surrounding vegetation in pre-monsoon conditions. The "urban" mean therefore includes both genuinely hot impervious surfaces and cooler industrial structures, compressing the apparent urban-rural thermal differential. The UHII trend ($+0.14^\circ\text{C}$) is the more informative figure here — it suggests a slow but real intensification of the urban thermal footprint relative to the vegetated background.

LST Uncertainty Quantification

In addition to mean LST, spatial standard deviation (σ) was computed for both epochs using a combined reducer (*ee.Reducer.mean().combine(ee.Reducer.stdDev())*). The σ reported here is the spatial standard deviation across all valid district pixels within a single composite — it describes the within-district variability of surface temperature at one point in time, not retrieval uncertainty. A district with high thermal heterogeneity (industrial cores, water bodies, forests) will inherently exhibit higher σ .

2015: Mean LST = **42.25 ± 2.98°C** | 2024: Mean LST = **48.70 ± 3.54°C**

The slight increase in σ between epochs ($2.98 \rightarrow 3.54^\circ\text{C}$) is consistent with growing intra-district thermal heterogeneity as built-up expansion concentrates heat in specific zones while vegetated areas remain cooler.

ERA5 Meteorological Detrending

ERA5-Land monthly aggregated reanalysis (ECMWF/ERA5_LAND/MONTHLY_AGGR) was used to extract mean 2m air temperature (temperature_2m) for the Bokaro district during the March–May observation windows of 2015 and 2024. ERA5-Land provides gridded reanalysis at

approximately 9 km resolution, sufficient to characterise regional synoptic temperature trends. Temperature values were extracted as district means after converting from Kelvin to Celsius.

ERA5 2m air temperature: 27.58°C (2015), 28.76°C (2024). Regional background warming: $\Delta T_{\text{ERA5}} = +1.19^\circ\text{C}$. The detrended LST anomaly is computed as:

$$\Delta T_{\text{detrended}} = \Delta T_{\text{LST}} - \Delta T_{\text{ERA5}} = 6.45 - 1.19 = \mathbf{5.26^\circ\text{C}}$$

Two limitations apply. First, ERA5 2m air temperature and Landsat LST are different physical quantities — one is a near-surface atmospheric temperature, the other is a radiometric surface emission temperature — and they are not directly interchangeable in a rigorous energy balance framework.

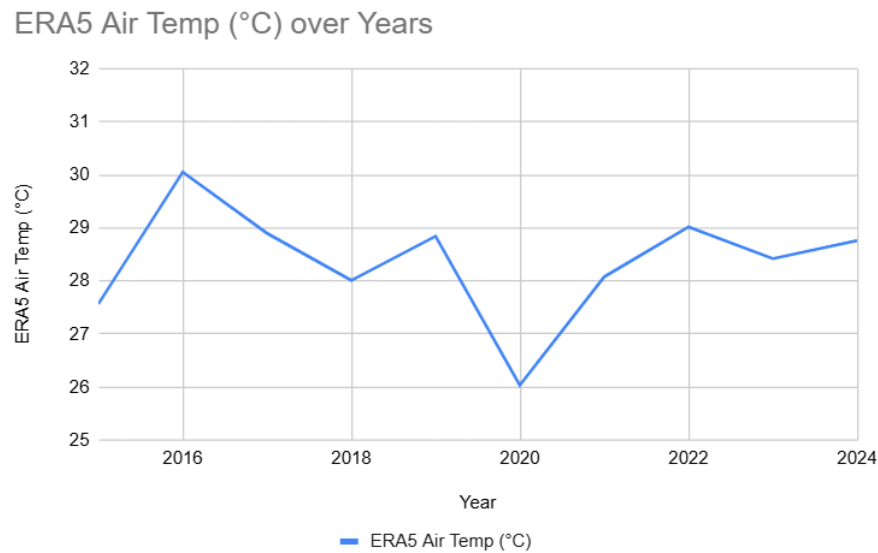


Figure 3: *ERA5 Air Temperature over the years*

The detrending is an approximation. Second, ERA5 at 9 km cannot resolve intra-district heterogeneity; it captures the average regional climate signal, not any urban-specific atmospheric warming.

Multi-Model Benchmarking

Dataset Preparation

A tabular feature dataset was exported from GEE by applying the `sampleRegions()` function to the 2024 composite image using the 2024 training polygon asset. The resulting CSV contains one row per sampled pixel, with nine spectral feature columns and a landcover class label. GEE metadata columns (*system:index* and *.geo*) were stripped prior to use.

The dataset was partitioned into 80% training and 20% test subsets using stratified random sampling. Stratification was applied to preserve the proportional representation of minority

classes—in particular, water bodies, which occupy a small fractional area of the district—across both subsets.

Importantly, all three local models were trained and evaluated on this same partitioned dataset, making their performance metrics directly comparable. The training set is itself derived from the same manually labeled polygons used for GEE classification, which introduces a risk of spatial autocorrelation: nearby pixels within a training polygon share spatially correlated spectral properties, and a random pixel-level train-test split does not guarantee spatial independence between subsets. This is a known limitation of pixel-based accuracy assessment and means that reported accuracy values likely overestimate generalization performance on independent held-out areas.

Model Architectures

Random Forest

A Random Forest ensemble with 100 trees (increased from the GEE setting of 50) was trained using gini impurity splitting and default hyperparameters. `n_jobs=-1` enabled parallel training across all available processor cores.

XGBoost

A gradient boosted tree ensemble (XGBoost `XGBClassifier`) with 200 estimators, maximum tree depth of 6, and learning rate of 0.1 was configured with the multiclass log-loss (`mlogloss`) evaluation metric. This architecture employs sequential residual correction rather than bagging, and is expected to converge to lower bias at the cost of higher sensitivity to data distribution.

Neural Network

A four-layer fully connected network was implemented in PyTorch with hidden dimensions [128 → 128 → 64 → 32]. Each hidden layer was followed by Batch Normalization, ReLU activation, and Dropout (rates: 0.3, 0.3, 0.2 at layers 1–3; no dropout at layer 4). The network was trained with Adam (`lr = 0.001`) using cross-entropy loss for 50 epochs, with batch size 64.

Performance Summary

All three models were evaluated on the held-out 20% test set. The results are summarized below:

	Random Forest	XGBoost	Neural Network (4 layer)
Overall Accuracy	0.98	0.98	0.98
Macro Avg F1 score	0.93	0.92	0.92
Weighted Avg F1 score	0.98	0.98	0.98
n_estimators / epochs	100 trees	200 trees	50 epochs

Table 3: *Benchmarking performance across three model architectures on the 20% stratified test partition.*

The near-identical performance across three structurally distinct learning algorithms is itself an informative result. It indicates that the classification problem, given the nine-feature spectral input, is not architecture-constrained: the feature-space separation between classes is sufficiently clean that any reasonably parameterized classifier can exploit it. The marginal F1 advantage of the Random Forest likely reflects its better handling of the minority water class, where ensemble diversity from bagging reduces variance on sparse training examples.

Scaled Spatial Inference on Full District GeoTIFF:

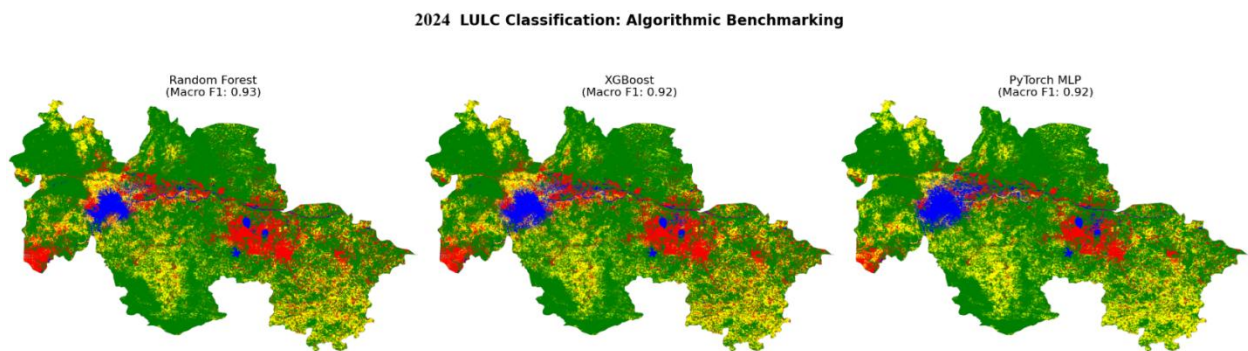


Figure 4: *Visualizing the prediction of each model*

The 2024 nine-band feature stack was exported from GEE as a Float32 GeoTIFF at 30 m resolution, covering the entire Bokaro district bounding box (approximately 6.72 million total pixel positions). This export was used to generate full-district LULC prediction maps from all three locally trained models.

[6.7M Raw Input Pixels] ---> [Boolean Masking Filter] ---> [3.5M Target District Pixels]

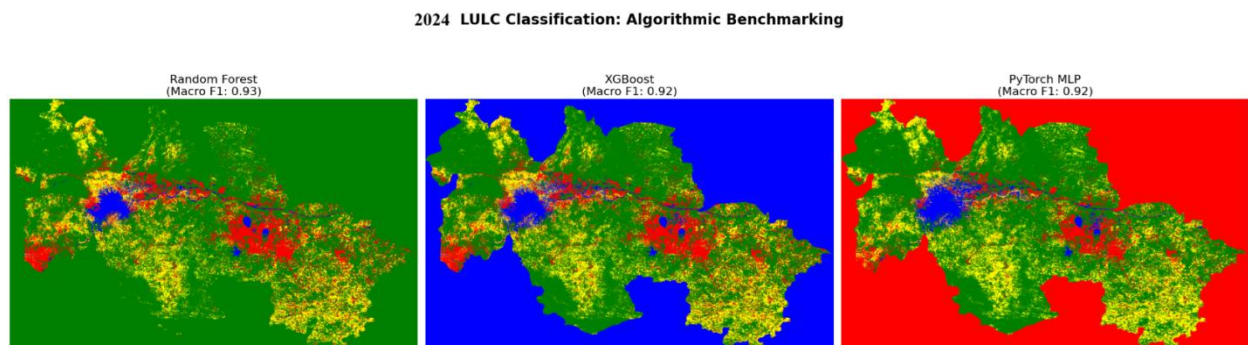


Figure 5: *Predictions without boolean masking (note how each model treats the NaN values outside district differently!)*

Mechanistic Interpretability via SHAP

Post-hoc SHAP attribution was applied to a stratified sample of 1,000 test instances to evaluate whether each model's internal classification logic aligns with physically grounded spectral understanding.

Tree-Based Models (RF and XGBoost)

Both tree-based models were analysed using *shap.TreeExplainer*, which computes exact Shapley values via the tree path.

For both models, MNDWI emerged as the highest-magnitude attributor for water-class discrimination. This result is physically coherent: MNDWI's large positive values for open water and strongly negative values for built-up surfaces provide a near-threshold separability that tree splits can exploit directly. The SHAP beeswarm plot for the built-up class confirmed that high MNDWI values suppress built-up probability strongly—functioning effectively as a structural veto. NDVI and NDBI played secondary attribution roles, consistent with their discriminative relevance for vegetation and impervious surface classes respectively.

PyTorch MLP

The neural network was analysed using *shap.DeepExplainer* with a background dataset of 100 randomly sampled training instances (sufficient for integration approximation at this model scale). SHAP values were returned as a 3D tensor requiring device-side conversion (*cpu().detach().numpy()*) before plotting.

The MLP showed markedly different attribution patterns from the tree-based models. NDVI and NDBI received low SHAP attributions across the sample, while the raw SWIR bands SR_B6 and SR_B7 dominated the feature importance landscape. This indicates that the network's four hidden layers learned to directly exploit the reflectance contrast between SWIR bands and other optical channels—effectively encoding a self-discovered representation analogous to NDBI—without relying on the pre-computed indices. SHAP also highlighted strong collinearity between NDBI and the raw SWIR stack: the network suppressed the redundant engineered feature in favour of the more information-dense raw inputs.

These contrasting attribution profiles are scientifically informative. They reveal that feature engineering decisions that are necessary for tree-based classifiers (which cannot implicitly compute band ratios) are partially redundant for neural architectures that can learn arbitrary nonlinear transformations. However, the engineered indices retain value as explicit domain-knowledge constraints that prevent physically implausible class assignments—particularly in low-data regimes where neural network generalisation cannot be guaranteed.



Figure 6: *Bee-Swarm plot of all three models*

Scaled Spatial Inference Map Generation

The final inference pipeline used the `rasterio` library to parse the massive 6.7 million-pixel multi-band GeoTIFF file. To prevent out-of-distribution class hallucinations—such as models predicting valid vegetation or water classes over empty boundary margins—a robust boolean masking layer was implemented.

This vector mask systematically identified and discarded all NaN and NoData perimeter values, narrowing the active inference space down to **3.5 million valid district pixels**. The filtered arrays were flattened, run through the three trained model pipelines, and reconstructed into 2D spatial matrices.

The resulting spatial arrays were visualized using `matplotlib`. The classified outputs across all three models were nearly identical, demonstrating that rigorous feature engineering cleanly separated the underlying LULC spectral classes, making the choice of model architecture secondary to data preparation.

Experimental Results and Discussion

Decadal LULC Transition Analysis (2015-2024)

The epoch-specific Random Forest classifiers produced spatially coherent LULC maps for both 2015 and 2024. The built-up class expanded from approximately 359.2 km² to 387.6 km², representing a net areal increment of 28.4 km² (+7.9%) over the decade. This is a significant absolute change relative to the total district area (~2,861 km²), reflecting steady increase in residential, commercial and industrial areas. Growth is spread out along the industrial periphery of Bokaro Steel City and along transit corridors radiating outward toward smaller townships.

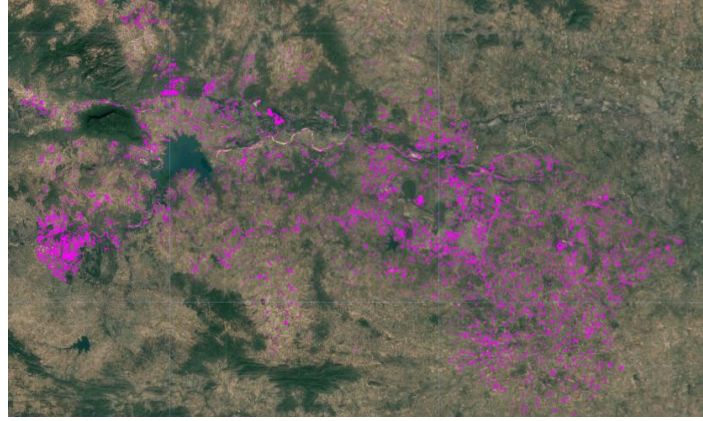


Figure 7: Areas of urban expansion between 2015 and 2024 (in pink)

Thermal Dynamics

Mean pre-monsoon LST across the Bokaro district rose from $42.25 \pm 2.98^{\circ}\text{C}$ in 2015 to $48.70 \pm 3.54^{\circ}\text{C}$ in 2024, a raw increase of 6.45°C . The spatial standard deviation increased modestly between epochs, consistent with growing intra-district thermal heterogeneity as built-up expansion concentrates heat in specific zones.

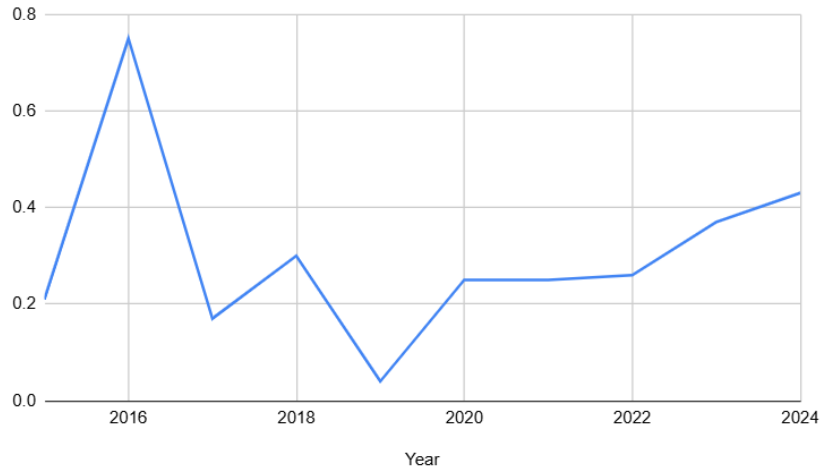


Figure 8: Line plot showing the change in UHII (i.e $LST_{urban} - LST_{rural}$)

After de-trending against ERA5-Land 2m air temperature (background regional warming of 1.19°C over the same period), the LULC-attributable surface warming estimate is 5.26°C . The stable rural vegetated control warmed by 6.31°C over the same period, nearly identical to the 6.45°C district-wide figure, which explains the small UHII values: the regional climate signal dominates the gross temperature change, while the urban-rural thermal differential is a secondary but growing signal (UHII: $+0.28^{\circ}\text{C}$ in 2015 to $+0.42^{\circ}\text{C}$ in 2024).

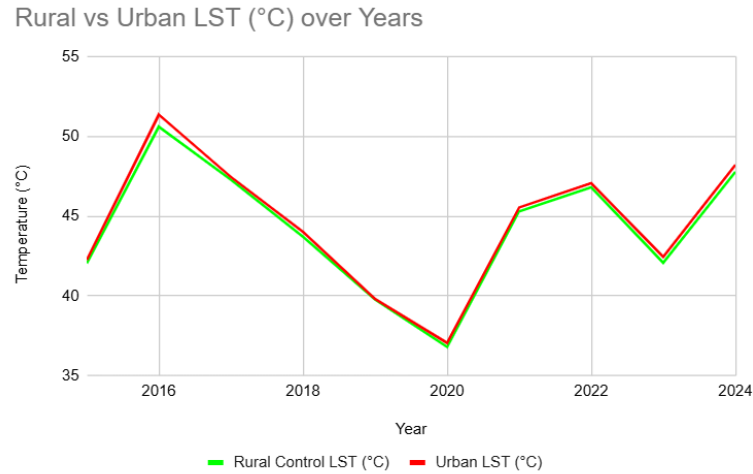


Figure 9: Line plots showing rural and urban LST over the years

The pixel-wise LST difference map shows strong spatial heterogeneity. The highest warming concentrations correspond to expanded impervious surfaces in the industrial periphery and road corridors; water bodies and dense forest patches show attenuated or negative anomalies depending on the year's hydrological conditions.

Caution is warranted when interpreting the absolute magnitude of these figures. The simplified emissivity model ($LSE = 0.004 \times FVC + 0.986$) introduces systematic uncertainty into absolute LST values, and ERA5 2m air temperature is not the same physical quantity as surface radiometric temperature. These results are best interpreted as directional indicators of thermal intensification rather than calibrated geophysical measurements.

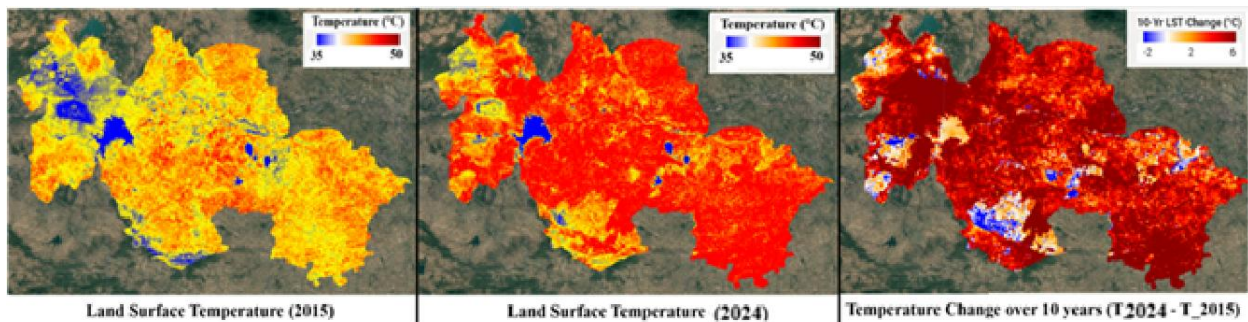


Figure 10: LST visualized on the district map

Algorithmic Convergence and Benchmarking

The convergence of all three classifiers to near-identical overall accuracy (0.98) and macro F1 (0.92–0.93) is the most reproducibility-relevant result of the benchmarking exercise. It demonstrates that once the feature space is well-engineered—spectral indices cleanly separating classes, LST excluded to prevent temporal drift—the downstream classifier architecture contributes marginally to performance. This has practical implications: for production UHI monitoring systems with limited compute budgets, a well-configured Random Forest with

appropriate feature engineering will perform comparably to a neural network, without the training complexity, hyperparameter sensitivity, or GPU infrastructure requirements.

Conversely, the SHAP analysis reveals that the three models are not interpretively equivalent despite their similar outputs. The neural network's ability to autonomously bypass engineered features and learn from raw spectral channels suggests that, with larger training datasets and deeper architectures, it may generalize better to spectrally ambiguous boundary classes—though this hypothesis is not directly testable within the current study's training data volume.

Future Work

To expand upon these findings, future iterations of this research will focus on the following core areas:

- **2015 Training data expansion:** The 2015 classifier's kappa of 0.756 is driven primarily by poor Bare Soil recall (29%). Adding more bare earth training polygons for 2015 and re-running the spatially independent validation is the clearest path to improving the baseline epoch's classification confidence.
- **Sentinel-2 integration:** Sentinel-2's 10 m spatial resolution in visible-NIR bands would enable finer-scale mapping of intra-urban thermal variability, capturing building-scale features invisible at Landsat's 30 m native resolution.
- **Spatial CNN:** The current pipeline treats each pixel as an independent vector, discarding spatial context. A convolutional approach over the multispectral image stack would allow the model to exploit patch-level texture and spatial arrangement, which are likely discriminative for distinguishing industrial structures from residential built-up areas at Landsat resolution.
- **Hydrological impact assessment:** The documented built-up expansion could be coupled with distributed hydrological models to estimate changes in impervious surface runoff and groundwater recharge within the Damodar River catchment.

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